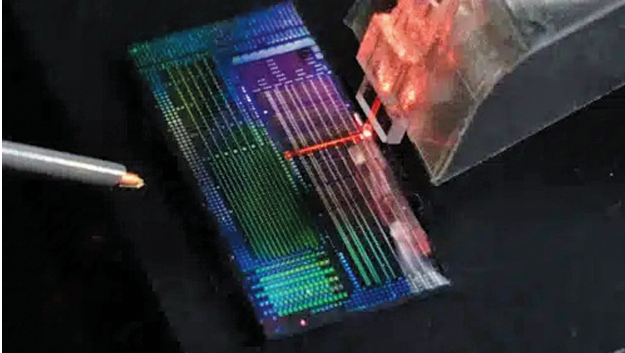


## Tiny chip could power large quantum computers

A miniature optical phase-modulator chip has been developed that could enable large-scale quantum computing by precisely controlling laser light with minimal power and heat. Researchers at the University of Colorado at Boulder designed the device to use microwave-frequency vibrations to shift laser phases, essential for qubit operations



Optical chip developed in the study with laser light from an optical fibre array (Credit: Jake Freedman)

in trapped-ion and neutral-atom systems, while reducing power consumption by roughly 80x compared with conventional modulators. Scalable chip fabrication further supports mass production, advancing photonic integration for quantum computing, sensing, and networking applications.

## AI steps into the laboratory

An AI agent has been developed that can autonomously design, run, and interpret real-world scientific experiments, advancing the push toward autonomous laboratories. Researchers at IIT Delhi, collaborating with teams from Denmark and Germany, created AILA (artificially intelligent lab assistant) to operate complex instruments like atomic force microscopes, adjusting parameters, executing measurements, and analysing results without human intervention. The system accelerates lab workflows and reduces dependence on expert operators, though safety controls remain essential before wider adoption.

## Ferroelectric capacitor for compact devices

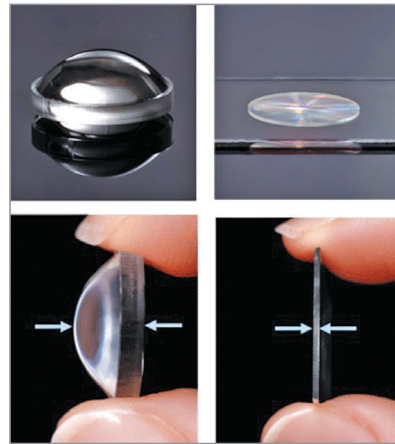
An ultrathin ferroelectric memory capacitor has been demonstrated, paving the way for compact, energy-efficient on-chip memory in mobile and IoT devices. A team at the School of Materials and Chemical Technology, Institute of Science, Tokyo, in collaboration with Canon ANELVA Corporation, built a 30nm thick stack of scandium-substituted aluminium nitride ( $(\text{Al}_{0.9}\text{Sc}_{0.1})\text{N}$ ) between platinum electrodes. Despite the minimal thickness, the capacitor retains strong switchable polarisation and can be directly integrated with logic circuits for ultracompact electronics.

## Next-gen cores for transformers

Next-generation transformer cores have been developed that reduce core losses by more than 50% at high frequencies, a crucial improvement for EV power electronics, high-frequency converters, and data centre power systems. Scientists from the National Institute for Materials Science (NIMS), Tohoku University, and AIST engineered soft magnetic iron-based amorphous ribbons and optimised thermal treatments to tailor nanostructures and magnetic domains. The resulting material enables compact, low-loss transformers and highly efficient power supplies for next-generation electrified systems.

## Meta-lens powers compact 3D wearable displays

An ultrathin meta-lens has been developed to enable compact, wearable 3D aerial displays by precisely controlling light focus based on wavelength. Kyocera Corporation engineered the metasurface lens to be under 1mm thick, integrating multiple optical functions and reducing reliance on bulky optics for lightweight, body-worn displays. By using colour-dependent focal control, the prototype projects floating images



Conventional refractive lens (left) and Kyocera's newly developed meta-lens (right) (Credit: <https://global.kyocera.com>)

with natural depth cues, creating immersive visuals. The innovation represents a step toward next-generation AR/VR wearables and other miniaturised optical systems.

## EV battery that doubles driving range

An anode-free lithium metal battery has been developed that could nearly double electric vehicle (EV) range without increasing pack size. A Korean team from POSTECH, KAIST, and Gyeongsang National University designed the battery to eliminate the conventional anode, achieving a volumetric energy density of about 1270Wh/L—nearly twice the  $\sim 650\text{Wh/L}$  of typical EV batteries. By packing more energy into the same space, the approach could enable longer trips per charge, though issues like lithium dendrite formation and long-term cycle stability remain research challenges.